

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix

Source: Our World in Data • Ember (2026) • Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas emissions come from burning fossil fuels for energy

5M+

premature deaths per year caused by air pollution from fossil fuel combustion

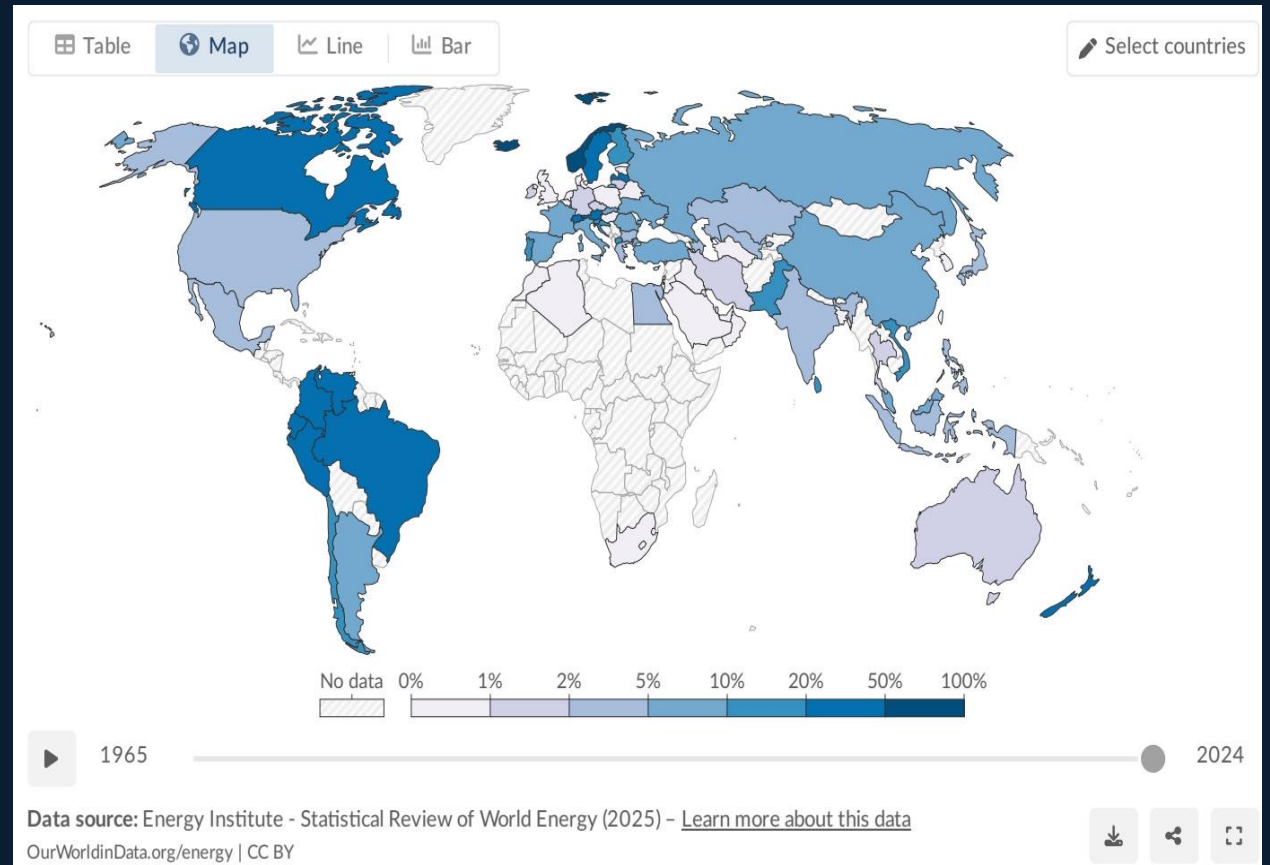
200+

years of fossil fuel dominance since the Industrial Revolution

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies including solar, wind, hydropower, geothermal, and bioenergy.

Renewables Now Supply ~14% of Global Primary Energy

- Approximately one-seventh of the world's primary energy comes from renewables (2024, substitution method)
- Primary energy = electricity + transport + heating combined
- Transport and heating remain harder to decarbonize — more reliant on oil and gas
- Norway, Brazil, and New Zealand derive large shares from renewables; Middle East nations remain below 5%



Renewable Electricity Generation Is Growing Rapidly

Led by Wind and Solar — Generation by Technology
(2024)



Technology Breakdown

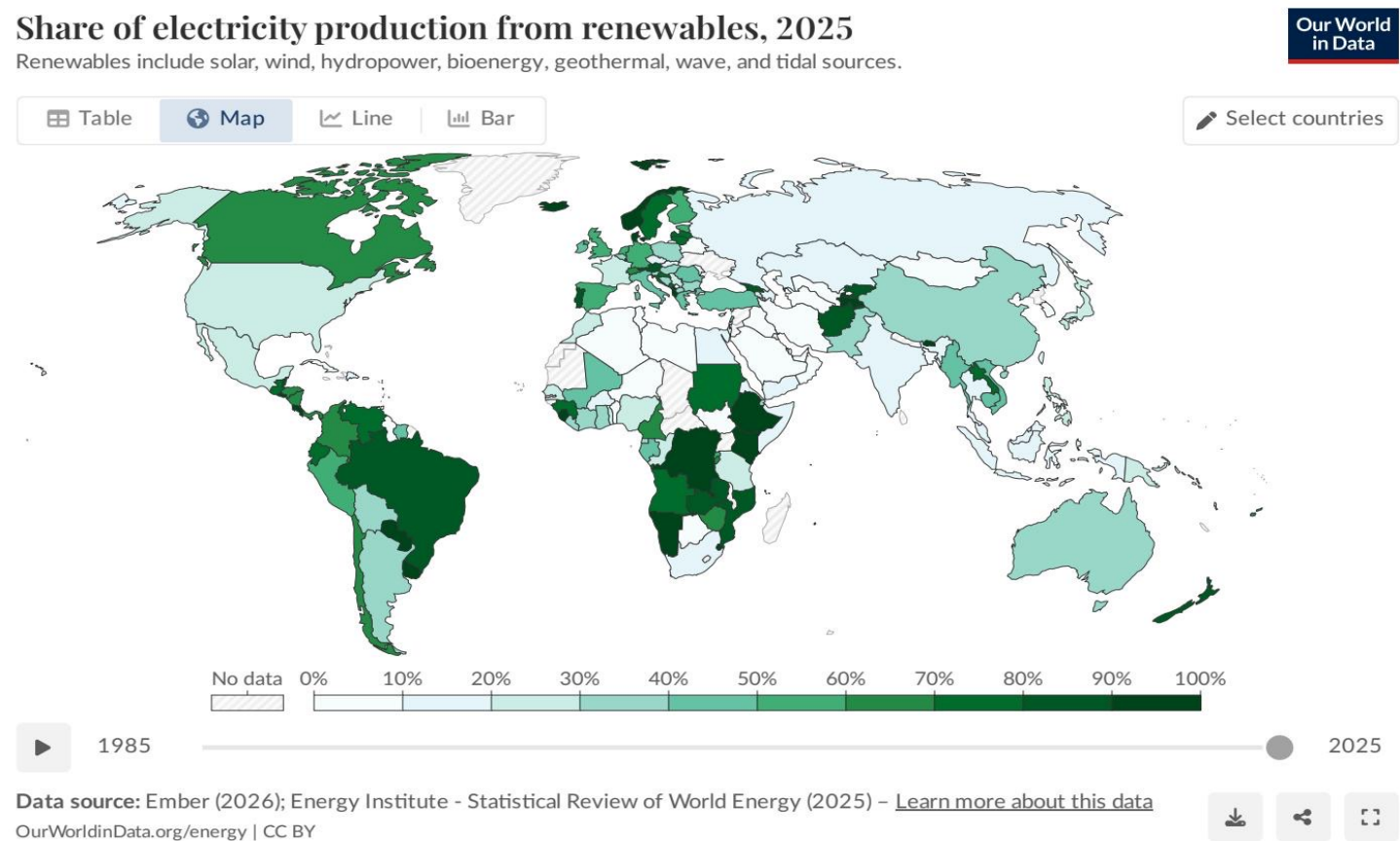
Hydropower	~4,500 TWh	~47%
Wind	~2,400 TWh	~25%
Solar	~2,000 TWh	~21%
Other	~700 TWh	~7%

Total **~9,600 TWh** **100%**

▶ Wind + Solar combined now nearly match
hydropower's output

▶ Solar is the fastest-growing source,
with dramatic growth since 2010

Almost One-Third of Global Electricity Now Comes from Renewables

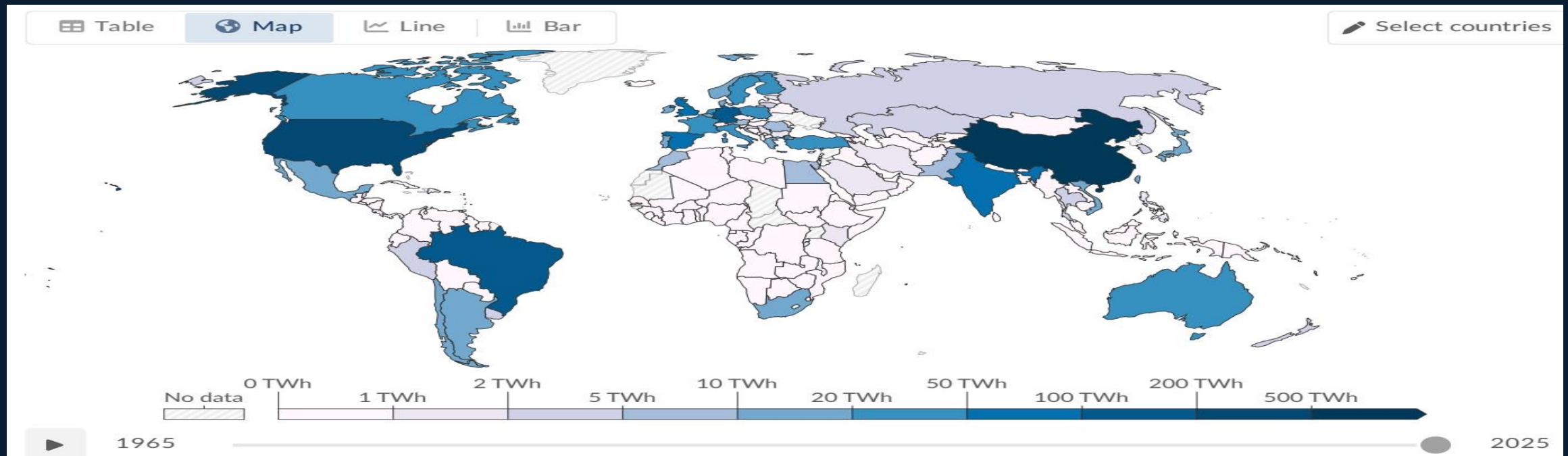


Country / Region	Renewable %
Norway	~98%
Brazil	~85%
Canada	~65%
Germany	~55%
China	~35%
India	~25%
World Average	~30%
Saudi Arabia	<2%

Leaders like Norway (98%) and Brazil (85%) contrast sharply with Saudi Arabia (<2%), reflecting vast regional disparities.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

- ~4,500 TWh generated in 2024 — roughly half of all renewable electricity
- Generating electricity at scale for over a century
- Top producers: China, Brazil, Canada



Wind and Solar Are the Fastest-Growing Energy Technologies in History

↑ WIND ENERGY

~2,400 TWh

~25% of renewable electricity

Major contributor in Europe, China, US

☀️ SOLAR ENERGY

~2,000 TWh

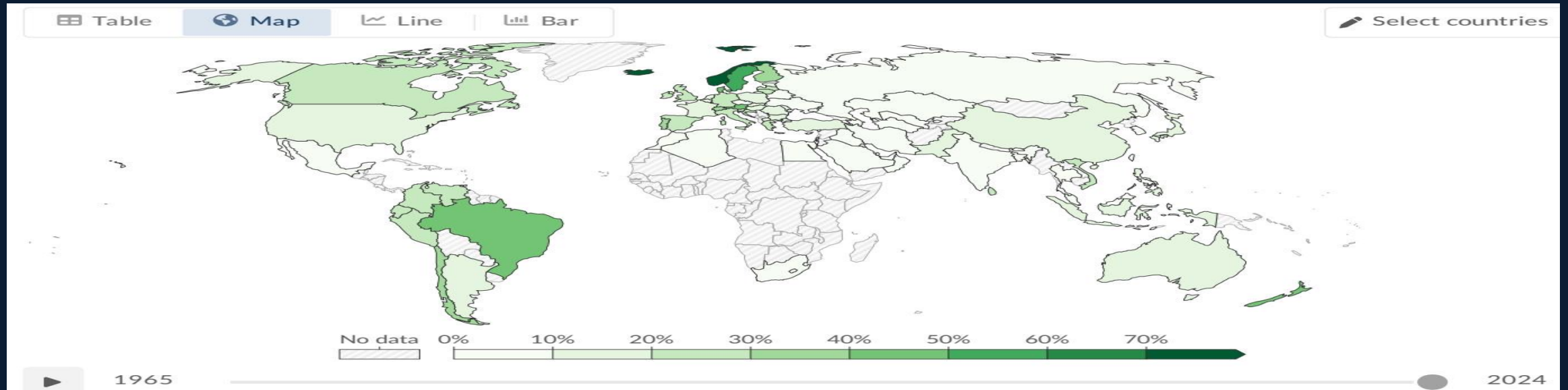
~21% of renewable electricity

Fastest-growing source since 2010

COMBINED

~4,400 TWh

Nearly matching hydropower's
~4,500 TWh output



Stark Regional Gaps in the Renewable Energy Transition

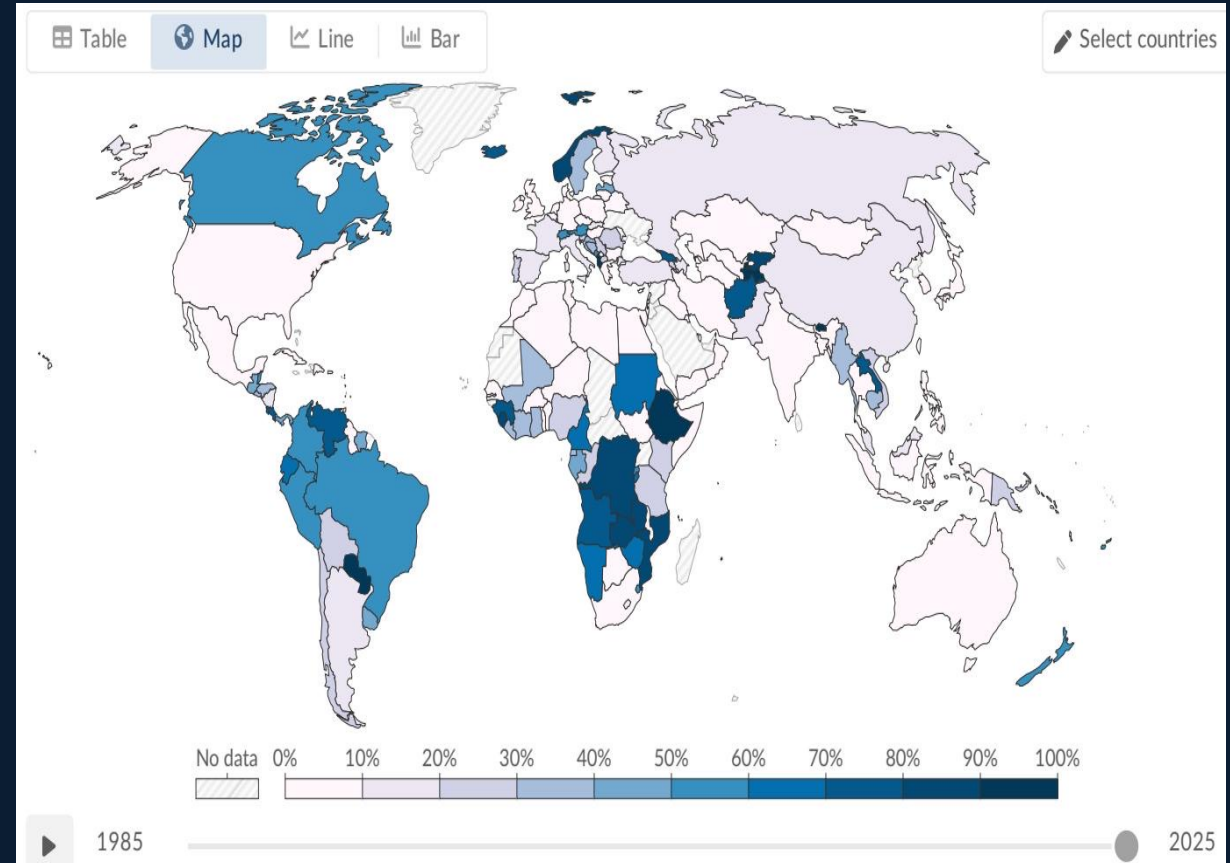
Leaders (60–98% renewable electricity)

- Scandinavia: abundant hydropower + heavy wind/solar investment
- Brazil & Canada: major hydropower capacity
- Denmark & Germany: pioneered wind and solar at scale

Lagging Regions (<10% renewable electricity)

- Sub-Saharan Africa: minimal renewable infrastructure
- Middle East: almost entirely dependent on fossil fuels
- Saudi Arabia: <2% renewable electricity share

The transition is highly uneven — geography, investment, and policy determine renewable energy outcomes.



Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14% of primary energy & ~30% of electricity

Renewables have reached a significant milestone, but remain far from sufficient for climate targets.

Hydropower still dominates at ~47% of renewable generation

Generating ~4,500 TWh, hydro leads — but its growth potential is limited compared to wind and solar.

Wind + Solar are the fastest-growing energy technologies

Combined ~4,400 TWh, they now nearly match hydropower. Solar has grown dramatically since 2010.

Stark regional disparities persist

Scandinavia and Latin America exceed 60–98%; much of the Middle East and Sub-Saharan Africa remain below 10%.

Transport and heating lag far behind electricity

These sectors remain heavily reliant on oil and gas, keeping overall primary energy share much lower than electricity.

The pace of the renewable energy transition must accelerate dramatically to meet global climate commitments.